Laplacian nonintegrality of three-prime ideal intersection graphs

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Abstract

We prove that the ideal intersection graph of $\mathbb{Z}_{p^a q^b r^c}$ is Laplacian nonintegral for any three distinct primes and every positive exponent triple. Vertices are the nonzero proper ideals; adjacency means nonzero intersection.

A six-support complement quotient C supplies the spectral obstruction. Symmetry and exact factor indices settle repeated exponents. For fully distinct exponents beyond the small-minimum cases, a positive quotient root below three reduces a hypothetical integer spectrum to endpoint one or endpoint two. Endpoint two supplies a further root in $(4, 5)$ at large minima; endpoint one reduces to two small-gap brackets and the largest-root interval

$$bc + a + b + c < \lambda_{\max}(C) < bc + a + b + c + 1$$

for real $2 \le a \le b \le c$, $b - a \ge 3$ and $c \ge a^2 - a$. The proof is computer-assisted: unbounded spectral arguments are combined with complete exact checks on finite domains of 325 pairs, 27 562 triples and 8 658 triples. The required coefficient identities and certificate specifications are supplied. Verification uses integer and rational arithmetic.

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1 Introduction

The ideal intersection graph records the intersection structure of the nonzero proper ideals of a ring [8]. For $\mathbb{Z}_n$, its isomorphism type depends on the prime exponents. We ask when its combinatorial Laplacian has only integer eigenvalues. Related ring graphs include Beck's coloring graph, zero-divisor graphs and annihilating-ideal graphs [3, 2, 4, 5, 1]. Their multiplication adjacency differs from the intersection convention used here. Laplacian and graph-spectral background is developed in [10, 13, 15, 9, 6]; equitable partitions and generalized joins reduce these graphs to small matrices [12, 7].

Nikandish [16, Theorems 13 and 30] gives a generalized-join spectrum formula and a Schur–Weyl reduction for equal prime exponents, with representation background in [11]. His nonintegrality result for $(pqr)^k$, $k \ge 1$ [16, Corollary 31], is extended here to every positive three-prime exponent triple (Theorem 9.3). The support-class graph model makes the prime values irrelevant. The main difficulty is therefore the unbounded exponent domain, including fully unequal exponents whose complement quotient can have integer low roots.

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Our main result is that every graph in this three-prime family is Laplacian nonintegral, for all positive integers a, b, c (Theorem 9.3). The multiplicatively related support weights permit symmetry reduction, exact factor arithmetic and real-parameter spectral separation across the unbounded exponent domain.

The proof separates three cases. Theorem 4.7 settles repeated entries, and Corollary 6.1 settles minima at most seven using endpoint signs, the uniform cutoff and complete finite certificates. For $8 \le a < b < c$, Theorem 6.3 puts a positive quotient root in $(0, 3)$. Integrality forces it to be one or two. Theorem 8.2 excludes two. At one, Lemma 9.2 handles middle gaps one and two; for larger gaps, Theorem 7.1 gives the exponent-sum bound needed by Theorem 9.1. This completes the classification without enumerating integer points on the endpoint-one surface.

The computer-assisted dependencies are stated where used and collected in Appendix A. The complete coefficient tables in the other appendices support the unbounded real arguments. The higher-prime nonsquarefree classification remains open. Further arithmetic obstructions, local equality-curve analysis and higher-prime families are retained in the accompanying research manuscript and repository [17]. Local Diophantine questions on that curve are separate from the spectral classification; their general background is in [14, 18].

2 The graph and its invariant subspaces

Write $n = \prod_{i=1}^{t} p_i^{k_i}$, where the primes are distinct and $k_i \ge 1$. Denote by G_n the graph whose vertices are the nonzero proper ideals of $\mathbb{Z}_n$, with distinct ideals adjacent if their intersection is nonzero. We use the combinatorial Laplacian $L = D - A$; the graph is *Laplacian integral* when every eigenvalue of L is an integer, including when the graph is disconnected.

The ideal $(\prod_i p_i^{\alpha_i})$ is encoded by $0 \le \alpha_i \le k_i$. We exclude $(0, \dots, 0)$, which encodes the whole ring, and $(k_1, \dots, k_t)$, which encodes the zero ideal. Ideal intersection is generated by the least common multiple. Therefore two distinct vertices α, β are adjacent exactly when

$$\max(\alpha_i, \beta_i) < k_i \quad \text{for some } i.$$

Equivalently, putting $S(\alpha) = \{i : \alpha_i < k_i\}$, adjacency is $S(\alpha) \cap S(\beta) \neq \varnothing$.

For nonempty $S \subseteq [t]$, the support class $\mathcal{C}_S$ is a clique of size $\prod_{i \in S} k_i$, except that the full support has size $u = \prod_i k_i - 1$. Edges between distinct support classes are complete or absent according as their supports intersect or are disjoint. The full support consists of universal vertices. Remove them and call the resulting graph H; thus $G_n = K_u \vee H$, with the evident interpretation for $u = 0$.

Lemma 2.1 (Lifting across the universal vertices)**.** *If $L(H)v = \lambda v$ and $\lambda \neq 0$, extension of v by zero on the u universal vertices is an eigenvector of $L(G_n)$ with eigenvalue $\lambda + u$.*

Proof. Since $L(H)$ is symmetric with zero row sums, $\lambda \neq 0$ implies $\sum v = 0$. At a vertex of H, its degree increases by u and all new neighbor values are zero. At a universal vertex the Laplacian action is $-\sum v = 0$. These are exactly the claimed eigenvector equations. □

For $(k_1, k_2, k_3) = (a, a, b)$, H has six support classes of sizes a, a, b, a^2, ab, ab on $1, 2, 3, 12, 13, 23$, respectively. The partition into

$$X = \mathcal{C}_1 \cup \mathcal{C}_2, \quad Y = \mathcal{C}_3, \quad Z = \mathcal{C}_{12}, \quad W = \mathcal{C}_{13} \cup \mathcal{C}_{23}$$

is equitable, and on functions constant on these cells the Laplacian acts by

$$B = \begin{pmatrix} a^2+ab & 0 & -a^2 & -ab \\ 0 & 2ab & 0 & -2ab \\ -2a & 0 & 2a+2ab & -2ab \\ -a & -b & -a^2 & a^2+a+b \end{pmatrix}. \tag{1}$$

For example, an X vertex has a^2 neighbors in Z and ab in W, while a W vertex has a, b, a^2 neighbors in X, Y, Z. Internal neighbors cancel for cell-constant functions. The cell sizes are $2a, b, a^2, 2ab$, and multiplying B on the left by their diagonal matrix gives a symmetric matrix. Thus B is similar to a real symmetric matrix.

The functions taking values $s, -s$ on $\mathcal{C}_1, \mathcal{C}_2$, $t, -t$ on $\mathcal{C}_{13}, \mathcal{C}_{23}$, and zero elsewhere form a second invariant subspace, with matrix

$$B_- = \begin{pmatrix} a^2+ab & -ab \\ -a & a^2+a+b+2ab \end{pmatrix}. \tag{2}$$

These functions have sum zero. Their eigenvalues therefore also lift by $u = a^2b - 1$, directly by the proof of Lemma 2.1.

Expanding (1) gives $\det(xI - B) = x f_{a,b}(x)$, where

$$\begin{aligned} f_{a,b}(x) &= x^3 - E_1x^2 + E_2x - E_3, \\ E_1 &= 2a^2 + 5ab + 3a + b, \\ E_2 &= a^4 + 7a^3b + 3a^3 + 8a^2b^2 + 11a^2b + 2a^2 + 3ab^2 + 2ab, \\ E_3 &= 2a^2b(a+b+1)(a^2+2ab+2a+b). \end{aligned} \tag{3}$$

Its roots are real and nonzero: reality follows from the similarity above, and $E_3 > 0$. Every root λ gives an eigenvalue $\lambda + u$ of G_n. A noninteger root of $f_{a,b}$ therefore proves nonintegrality.

3 Structure of the proof

The complement quotient explains the endpoint alternatives in the three-prime classification. Its low-root bound localizes a possible integer spectrum to two endpoint surfaces, whose geometric bounds lead to different spectral obstructions. We summarize this mechanism before giving the component proofs.

3.1 Localization and the sharp threshold

Let C be the complement Laplacian quotient on the six proper nonempty supports, of weights a, b, c, ab, ac, bc, and put $h_C(x) = \det(xI - C)/x$. The off-diagonal entry from S to T is $-w_T$ when $S \cap T$ is empty, and zero otherwise; the diagonal is the sum of these neighbor weights. The weighted symmetric quotient is positive semidefinite with a simple zero eigenvalue, since the disjointness graph on these supports is connected.

For positive integer exponents, every nonzero complement quotient root must be an integer if G_n is Laplacian integral: reflection across the complement and the universal-vertex lift transfer every noninteger root to a noninteger graph eigenvalue. Theorem 6.3 gives a positive quotient root below three for every ordered positive real triple. Hence the smallest positive root also lies in $(0, 3)$, and an integral graph spectrum requires

$$h_C(1) = 0 \qquad \text{or} \qquad h_C(2) = 0.$$

These are necessary conditions; the following arguments exclude integer spectra on each surface by a further root or an exponent-root bracket.

Proposition 3.1. *The uniform threshold three is optimal even for integer exponents. For $a = b = c = m \geq 1$, the smallest positive quotient root satisfies*

$$\mu_- = \frac{m^2 + 4m - m\sqrt{(m+4)^2 - 12}}{2} \in \left(\frac{3m}{m+4}, 3\right), \qquad \lim_{m\to\infty} \mu_- = 3.$$

Proof. Direct substitution in the quotient gives

$$h_{m,m,m}(x) = (x - m^2 - m)\big[x^2 - (m^2 + 4m)x + 3m^2\big]^2.$$

For the quadratic q_m, we have $q_m(0) > 0$ and $q_m(m) = -m^3 < 0$. Its smaller root therefore lies in $(0, m)$ and is below the other three positive roots, counted with multiplicity. Also $q_m(3) = 9 - 12m < 0$, so $\mu_- < 3$. Rationalization gives

$$\mu_- = \frac{6m}{m + 4 + \sqrt{(m+4)^2 - 12}} > \frac{3m}{m+4}.$$

Dividing the denominator by m proves the limit, including along positive integers. Thus no smaller constant is a uniform upper bound. □

3.2 The endpoint surfaces and their scales

After repeated exponents and minima at most seven have been settled, the remaining classification domain is $8 \leq a < b < c$. In this domain, Lemma 8.1 shows that

$$h_C(2) = 0 \quad \Longrightarrow \quad b < 2a - 2, \qquad c < \frac{9a}{4} - 8.$$

Both larger exponents are bounded multiples of the minimum. The rectangle is finite at each fixed integer a, but a is unbounded. Theorem 8.2 supplies a root in $(4, 5)$ for $a \geq 40$; the complete 8 658-triple certificate handles $8 \leq a \leq 39$. The real tail argument and the finite base together close this alternative.

Endpoint one has a different necessary scale. For real $2 \leq a \leq b \leq c$, Theorem 7.1 gives

$$h_C(1) = 0 \quad \Longrightarrow \quad b + c \geq 2a^2 - 2a + 1, \qquad c \geq a^2 - a + \tfrac{1}{2}.$$

The second inequality follows from $b \leq c$. This quadratic lower bound supplies the hypothesis of the largest-root interval in Theorem 9.1 whenever $b - a \geq 3$. For gaps one and two, Lemma 9.2 instead places the only possible exponent-root above b between consecutive integers. This route settles the spectrum without enumerating integer points on the endpoint-one surface.

3.3 The fixed-gap endpoint-one expansion

The small-gap anchors also have an asymptotic interpretation. The following result concerns the exponent-root c, with the gap fixed; it is supplementary to the exact brackets used in the classification.

Proposition 3.2. *Fix a real $d \geq 1$ and put $b = a + d$. For all sufficiently large real a, the equation $h_C(1) = 0$, regarded as a cubic in c, has exactly one root c^* above b. As $a \to \infty$ with d fixed,*

$$c^* = 2a^2 + (2d-3)a - \frac{(d+2)(2d-1)}{2} + \frac{d^2}{2a} + \frac{d^2(d+1)(2d-3)}{4a^2} + O_d(a^{-3}). \tag{4}$$

The remainder constant and the required minimum may depend on d.

Proof. Write $Q_{a,d}(c) = h_C(1)$ at $b = a + d$. The cubic coefficients are displayed in Lemma 9.2; their identity holds for arbitrary d. Set $t = 1/a$, $y = t^2 c$ and

$$F_d(t, y) = t^8 Q_{1/t,d}(y/t^2).$$

This is a polynomial in t, y, with

$$F_d(0, y) = 2y^2(y-2), \qquad \partial_y F_d(0, 2) = 8.$$

The analytic implicit-function theorem gives a unique real analytic branch $y_d(t)$ near $t = 0$ with $y_d(0) = 2$. Substitution and comparison of successive powers of t give

$$y_d(t) = 2 + (2d-3)t - \frac{(d+2)(2d-1)}{2}t^2 + \frac{d^2}{2}t^3 + \frac{d^2(d+1)(2d-3)}{4}t^4 + O_d(t^5).$$

Multiplication by t^{-2} proves (4). For fixed d and sufficiently large a, the cubic has positive leading coefficient and $Q_{a,d}(0) > 0$, whereas

$$Q_{a,d}(a + d) = -4a^6 + O_d(a^5) < 0.$$

It therefore has a root in each of $(-\infty, 0)$ and $(0, b)$. The analytic branch has $c^* \sim 2a^2 > b$ and supplies the third root. These distinct roots exhaust the cubic degree, proving uniqueness above b. □

For $d = 1, 2$, the polynomial parts of this expansion are respectively $2a^2 - a - 2$ and $2a^2 + a - 6$, followed by a positive term of order a^{-1}. The exact unit brackets in Lemma 9.2 hold for all real $a \geq 2$. The expansion here is for fixed d; it supplies no uniform error bound when d grows with a. The largest-root argument handles every $d \geq 3$ in the classification using the quadratic lower bound on c.

4 Repeated exponents

Coordinate interchange gives the two invariant blocks above. The antisymmetric discriminant handles most parameters; exact factor indices and the symmetric cubic settle every square-discriminant exception.

4.1 Two unit exponents

Theorem 4.1. *For distinct primes p, q, r and every $k \geq 1$, the graph G_{pqr^k} has the irrational eigenvalues*

$$\frac{6k+1 \pm \sqrt{(2k+2)^2-3}}{2}.$$

In particular, every permutation of the exponent vector $(1,1,k)$ gives a nonintegral Laplacian spectrum.

Proof. The support classes $1, 2$ have size 1, and $13, 23$ have size k. Consider functions with values $s, -s$ on the first two classes, $t, -t$ on the latter two, and zero elsewhere. A vertex of support 1 has $2k$ neighbors and Laplacian value $2ks - kt$. A vertex of support 13 has degree $4k$; after subtracting the same-class and opposite-class values its Laplacian value is $-s + (4k+1)t$. The swapped classes give their negatives, and all other classes give zero by cancellation. The restricted matrix of the full graph is therefore

$$B_k = \begin{pmatrix} 2k & -k \\ -1 & 4k+1 \end{pmatrix}.$$

Its characteristic polynomial is $x^2 - (6k+1)x + 8k^2 + k$, with discriminant $(2k+2)^2 - 3$. Since

$$(2k+1)^2 < (2k+2)^2 - 3 < (2k+2)^2$$

for $k \geq 1$, the two roots are irrational. Relabeling prime coordinates preserves the graph. □

4.2 The boundary family

This is the index-zero case in the factor reduction below.

Theorem 4.2. *For every $a \geq 2$, put $C = (a-1)(2a-1)$. For distinct primes p, q, r, the graph $G_{p^a q^a r^C}$ is not Laplacian integral, although the antisymmetric block (2) has integer eigenvalues.*

Proof. Specialize $f_{a,b}$ from (3) to $b = C$. For $a = 2$ it is $x^3 - 47x^2 + 702x - 3312$. Its values at $0, 1, 2, 3, 4$ modulo 5 are $3, 4, 2, 3, 3$. Thus it has no root in $\mathbb{F}_5$ and is irreducible over $\mathbb{Q}$; its real roots are noninteger. Lemma 2.1 gives noninteger graph eigenvalues.

For $a \geq 3$ put $m = 2a^3 - 2a^2 + 2a - 1$. Expansion gives

$$\begin{aligned} f_{a,C}(m) &= 2(5a^4 - 10a^3 + 7a^2 - 1), \\ f_{a,C}(m-1) &= -2(a-1)(a+2)(2a^4 - 7a^3 + 7a^2 - a - 3). \end{aligned}$$

The first value is positive because its parenthesis is $5a^3(a-2)+7a^2-1$. Writing $z = a-3 \geq 0$, the last factor in the second value is $2z^4+17z^3+52z^2+68z+30 > 0$. Consequently $f_{a,C}(m-1) < 0 < f_{a,C}(m)$, so a real root lies strictly between $m-1$ and m. Its integer shift by $u = a^2C - 1$ is noninteger.

The characteristic discriminant of B_- is $D = (a+1)^2b^2 + 2a(3a+1)b + a^2$. At $b = C$ it equals $(2a^3 - a^2 + a - 1)^2$, and the two eigenvalues are $4a^3 - 3a^2 + a$ and $2a^3 - 2a^2 + 1$. Thus the obstruction in this family comes from the symmetric cubic. □

4.3 Exact factor pairs and the first indices

Fix $a, b \geq 1$, and put

$$h = a+1, \quad C = (a-1)(2a-1), \quad M = a^3(2a+1), \quad A = h^2b + a(3a+1).$$

The characteristic discriminant of (2) is

$$D = h^2b^2 + 2a(3a+1)b + a^2. \tag{5}$$

If D is not a square, this block has irrational eigenvalues, which remain irrational after the integer shift $a^2b - 1$.

Proposition 4.3 (Exact divisor reduction)**.** *For fixed $a \geq 1$, the positive integers b for which D is a square are exactly those obtained from positive factor pairs $d \leq e$ of M such that*

$$b = \frac{d + e - a(3a+1)}{h^2} \in \mathbb{Z}_{\geq 1}, \qquad s = \frac{e-d}{h} \in \mathbb{Z}_{\geq 0}. \tag{6}$$

In that case $D = s^2$.

Proof. The polynomial identity

$$A^2 - h^2 D = 4M \tag{7}$$

holds for arbitrary a, b. If $D = s^2$ with $s \geq 0$, then $A - hs$ and $A + hs$ are positive: their product is positive and their sum is $2A > 0$. They have the same parity; since their product is divisible by 4, both are even. Write $A - hs = 2d$ and $A + hs = 2e$. Then $de = M$, $d + e = A$ and $e - d = hs$, giving (6). Conversely those conditions yield $A = d + e$ and $hs = e - d$. Using $(d+e)^2 - (e-d)^2 = 4de = 4M$ in (7) gives $D = s^2$. □

The divisor set is finite for every fixed a; this criterion has no truncation in b. It gives first a quadratic bound.

Corollary 4.4. *If D is a square then $b \leq C$. Consequently every $b > C$ gives a nonintegral Laplacian spectrum. For $a = 1$, every $b \geq 1$ is covered.*

Proof. Since $(d-1)(e-1) \geq 0$, $A = d + e \leq M + 1$. The identity

$$M + 1 - a(3a+1) = h^2 C$$

then gives $b \leq C$. The nonsquare case is covered by the antisymmetric block. At $a = 1$, $C = 0$. □

Lemma 4.5. *Every admissible factor pair in Proposition 4.3 satisfies $d \equiv e \equiv 1 \pmod{h}$.*

Proof. From $e - d = hs$ and $d + e = A$, reduction modulo h gives $e \equiv d$ and $2d \equiv 2$, since $a(3a+1) \equiv 2 \pmod{h}$. Also

$$(d-1)(e-1) = M - A + 1 = h^2(C - b). \tag{8}$$

If h is odd, the congruence $2(d-1) \equiv 0$ gives the assertion. If h is even, the alternative to $d \equiv 1$ is $d - 1 \equiv h/2 \pmod{h}$. Then $d - 1$ and $e - 1$ would both equal $h/2$ times an odd integer. Their product divided by h^2 would be an odd integer divided by 4, contradicting (8). This excludes the alternative for either parity of a. □

The first possible value after $d = 1$ is $d = a + 2$. Since d divides M, the identity

$$M = (a+2)(2a^3 - 3a^2 + 6a - 12) + 24$$

forces $a + 2 \mid 24$. With $a \geq 2$ this leaves $a = 2, 4, 6, 10, 22$. Substitution into (6) gives $b = 0, 2, 5, 12, 35$, respectively. Excluding $b = 0$, the complete positive-b list and its cubic certificates are:

(a,b)	$f_{a,b}(x)$	Root-free prime
$(4,2)$	$x^3-86x^2+2304x-18816$	5
$(6,5)$	$x^3-245x^2+19266x-488160$	7
$(10,12)$	$x^3-842x^2+230160x-20534400$	13
$(22,35)$	$x^3-4919x^2+7887858x-4132479120$	17

Each polynomial has no root modulo its indicated prime. Its monic cubic is therefore irreducible over $\mathbb{Q}$, and its real nonzero roots lift to noninteger graph eigenvalues. Thus every possible $d=a+2$ pair is settled, over all a, by this finite divisor argument.

4.4 A finite reduction for each factor index

For a nonboundary square-discriminant pair, write $d=1+hj$, $j\geq 1$. The following criterion fixes j instead of a.

Proposition 4.6. *Fix $j\geq 1$ and set $K_j=(j+1)^3(j+2)$. Every admissible square-discriminant pair of index j is obtained from a positive divisor d of K_j with $a=(d-1)/j-1\in\mathbb{Z}_{\geq 2}$. Set $e=M/d$, $v=(e-1)/(a+1)$, and retain exactly those candidates with $d\leq e$ and $b=C-jv\geq 1$. Then e,v,b are integers and $D=(v-j)^2$. Conversely every retained candidate is admissible.*

Proof. Modulo $d=j(a+1)+1$ we have $ja\equiv -(j+1)$, whence

$$j^4M=(ja)^3j(2a+1)\equiv (j+1)^3(j+2)=K_j \pmod{d}.$$

Since $\gcd(d,j)=1$, we have $d\mid M$ if and only if $d\mid K_j$. Thus e is integral. Since $M\equiv d\equiv 1 \pmod{h}$, also $e\equiv 1 \pmod{h}$, so v is integral. The identity

$$M=1+h(3a-2)+h^2C=(1+hj)(1+hv)$$

gives $j+v+hjv=3a-2+hC$. Defining $b=C-jv$, we obtain $d+e=h^2b+a(3a+1)$ and $e-d=h(v-j)$. The ordering condition gives $v-j\geq 0$; Proposition 4.3 now proves admissibility and $D=(v-j)^2$. The reverse direction follows by applying these identities to any admissible index-j pair. □

This is a complete finite problem for each fixed j, without a truncation in a. Section 4 combines the first three such problems with a uniform argument for all later indices. For $j=2$, $K_2=108$ and $d=2a+3$ is an odd divisor of 108 with $d\geq 7$. The only candidates are $d=9,27$, giving $a=3,12$. The first has $e=21, v=5, b=0$ and is excluded. The second has $e=1600, v=123, b=7$ and $\sqrt{D}=121$. Thus $(12,7)$ is the only positive-b case with $j=2$.

Its symmetric cubic is $f_{12,7}(x)=x^3-751x^2+180348x-13829760$. Its values at all residues modulo 13 are $4,7,9,3,8,4,10,6,11,5,7,10,7$. As a monic cubic with no root modulo 13, it is irreducible over $\mathbb{Q}$. Its real nonzero roots lift to noninteger graph eigenvalues, settling the entire index-2 case.

4.5 Completion by a cubic interval

Theorem 4.7. *For distinct primes p,q,r and all integers $a,b\geq 1$, the graph $G_{p^aq^ar^b}$ is not Laplacian integral. The same conclusion holds for every permutation of the exponent vector (a,a,b).*

The arithmetic reduction and a uniform sign obstruction for the cubic combine to exhaust the family. We first settle one last finite factor level.

Lemma 4.8. *Every square-discriminant pair with index $j = 3$ gives a nonintegral Laplacian spectrum.*

Proof. Proposition 4.6 gives $d \mid K_3 = 320$ and $d = 3a + 4$. For $a \geq 2$, the complete divisor list of this shape is $10, 16, 40, 64, 160$. The resulting values are

d	a	e	v	b	Outcome
10	2	4	1	0	$d > e$
16	4	36	7	0	$b = 0$
40	12	1080	83	4	Root-free modulo 5
64	20	5125	244	9	Root-free modulo 7
160	52	92274	1741	30	Root-free modulo 7

The three positive-b cubics are

$$\begin{aligned} f_{12,4}(x) &= x^3 - 568x^2 + 100032x - 5248512, \\ f_{20,9}(x) &= x^3 - 1769x^2 + 992820x - 174744000, \\ f_{52,30}(x) &= x^3 - 13394x^2 + 57771168x - 80229951360. \end{aligned}$$

Their values at all residues of the indicated primes are, respectively, $(3, 3, 3, 4, 2)$, $(3, 2, 4, 1, 6, 4, 1)$ and $(1, 6, 4, 1, 3, 2, 4)$. Each monic cubic is therefore irreducible over $\mathbb{Q}$ and has noninteger real roots. Section 2 supplies the lift to noninteger graph eigenvalues. □

Lemma 4.9 (A uniform consecutive-integer interval)**.** *For $a, b \geq 1$, put*

$$L(a) = \frac{a^3 + 2a^2 + 2a + 1}{a(2a + 1)}.$$

If $b < L(a)$, then $f_{a,b}$ has a root strictly between $2ab - 1$ and $2ab$.

Proof. Substitution in (3) gives

$$\begin{aligned} f_{a,b}(2ab) &= 2a^2b^2 > 0, \\ f_{a,b}(2ab - 1) &= -(a + b + 1)\big(a^3 + 2a^2 + 2a + 1 - a(2a + 1)b\big). \end{aligned}$$

The second value is strictly negative when $b < L(a)$. The intermediate value theorem gives the asserted noninteger, nonzero root. □

Proof of Theorem 4.7. For $a = 1$, Corollary 4.4 settles every positive b. For $a \geq 2$, nonsquare D gives irrational antisymmetric eigenvalues. When D is square, index $j = 0$ is Theorem 4.2, and indices $j = 1, 2$ are settled in Section 4. Lemma 4.8 settles $j = 3$.

It remains to consider $j \geq 4$. Then $d \geq Q = 4a + 5$, and $e \geq d \geq Q$ implies

$$\frac{M}{Q} + Q - (d + e) = (d - Q)\left(\frac{e}{Q} - 1\right) \geq 0.$$

Since $d + e = (a + 1)^2 b + a(3a + 1)$, we obtain

$$b \leq B_4(a) = \frac{M/Q + Q - a(3a + 1)}{(a + 1)^2} = \frac{(a - 5)(2a - 5)}{4a + 5}.$$

For every $a \geq 2$,

$$L(a) - B_4(a) = \frac{41a^3 - 17a^2 - 11a + 5}{a(2a+1)(4a+5)} > 0.$$

Indeed, with $z = a-2 \geq 0$, the numerator becomes $41z^3+229z^2+413z+243$. Thus $b \leq B_4(a) < L(a)$, and Lemma 4.9 gives a noninteger cubic root, which lifts across the universal vertices. These cases exhaust all positive a, b. Permuting the prime factors preserves the graph's isomorphism type. □

5 The complement quotient and small minima

For unequal exponents the complement quotient retains all six support classes. Its positive roots reflect to graph eigenvalues by the complement identity and the universal-vertex lift.

5.1 A unit exponent

Theorem 5.1. *For distinct primes p, q, r and every $b, c \geq 1$, the graph $G_{pq^br^c}$ is not Laplacian integral. The same holds for permutations of the exponent vector $(1, b, c)$.*

The equal-exponent decomposition is replaced by a complement identity on all six support classes. For arbitrary $a, b, c \geq 1$, remove the $u = abc - 1$ universal vertices and denote the remaining graph by H. In the order $1, 2, 3, 12, 13, 23$, its class-size vector is $w = (a, b, c, ab, ac, bc)^T$ and its vertex count is $N = a + b + c + ab + ac + bc$. Disjoint supports are adjacent in the complement $\overline{H}$, whose cell-constant Laplacian quotient is

$$C = \begin{pmatrix} b+c+bc & -b & -c & 0 & 0 & -bc \\ -a & a+c+ac & -c & 0 & -ac & 0 \\ -a & -b & a+b+ab & -ab & 0 & 0 \\ 0 & 0 & -c & c & 0 & 0 \\ 0 & -b & 0 & 0 & b & 0 \\ -a & 0 & 0 & 0 & 0 & a \end{pmatrix}.$$

Multiplication on the left by $\mathrm{diag}(w)$ makes C symmetric. Moreover $C\mathbf{1} = 0$ and $w^TC = 0$. If B is the quotient of H, the complement relation gives

$$B + C = NI - \mathbf{1}w^T. \tag{9}$$

Thus an eigenvector $Cv = \mu v$ with $\mu \neq 0$ satisfies $w^Tv = 0$ and $Bv = (N - \mu)v$. When $N - \mu \neq 0$, Lemma 2.1 lifts it to an eigenvalue $N + u - \mu$ of G_n.

Proof of Theorem 5.1. Write $\det(xI - C) = xh_{a,b,c}(x)$. Direct determinant expansion yields

$$h_{a,b,c}(0) = -abc(a+b+c)N,$$
$$h_{a,b,c}(a) = -a^2b^2c^2(a^2 + 2a - b - c).$$

At $a = 1$, the first value is strictly negative and

$$h_{1,b,c}(1) = b^2c^2(b + c - 3).$$

When $b + c > 3$, a real root μ lies strictly between 0 and 1. Equation (9) and the universal-vertex lift produce a Laplacian eigenvalue $V - \mu \in (V - 1, V)$, where $V = N + u$ is the integer vertex count. This eigenvalue is noninteger. When $b + c \leq 3$, the positive pairs are $(1, 1), (1, 2), (2, 1)$; Theorem 4.1 settles the corresponding permutations of $(1, 1, k)$. Permuting the prime factors preserves the graph's isomorphism type. □

5.2 A minimum exponent of two

Theorem 5.2. *For distinct primes p, q, r and all $b, c \geq 1$, the graph $G_{p^2q^br^c}$ is not Laplacian integral. Consequently every three-prime exponent vector with minimum entry at most two gives a nonintegral graph.*

Proof. Use the complement quotient and quintic from Section 5. For $a = 2$, $h_{2,b,c}(0) < 0$. Putting $s = b + c$ and $t = bc$ gives

$$h_{2,b,c}(1) = (s - 13)t^2 - (2s + 6)t + 3s^3 + s^2 - 3s - 1.$$

By symmetry order $b \leq c$. A unit exponent is covered by Theorem 5.1; if $b = 2$ or $b = c$, use Theorem 4.7. It remains to consider $3 \leq b < c$.

When $b = 3$ and $c = 5 + v$, $v \geq 0$, the exact expansion is

$$h_{2,3,5+v}(1) = 12v^3 + 112v^2 + 268v + 120 > 0.$$

When $b \geq 4$ and $c > b$, write $b = 4 + u$ and $c = 5 + u + v$, with $u, v \geq 0$. Then

$$\begin{aligned} h_{2,4+u,5+u+v}(1) = {} & (u^2 + 8u + 19)v^3 \\ & + (4u^3 + 38u^2 + 128u + 170)v^2 \\ & + (5u^4 + 62u^3 + 271u^2 + 502u + 368)v \\ & + 2u^5 + 32u^4 + 190u^3 + 504u^2 + 552u + 160 > 0. \end{aligned}$$

All coefficients are positive. In either region the opposite signs at 0 and 1 give a root $\mu \in (0, 1)$ and, by (9) and the universal-vertex lift, a noninteger graph eigenvalue in $(V - 1, V)$.

The only remaining pair is $(b, c) = (3, 4)$. Here

$$h_{2,3,4}(x) = x^5 - 53x^4 + 953x^3 - 6523x^2 + 13134x - 7560,$$

and $h_{2,3,4}(1) = -48$ while $h_{2,3,4}(3/2) = 13437/32 > 0$. A root lies in $(1, 3/2)$. Since $V = 58$, its lifted graph eigenvalue lies in $(113/2, 57)$, an interval containing no integer. This exhausts all positive b, c. Combining with Theorem 5.1 proves the minimum-entry assertion. □

5.3 The uniform cutoff and minimum three

Proposition 5.3. *For distinct primes p, q, r and positive exponents ordered as $2 \leq a \leq b \leq c$, the graph $G_{p^aq^br^c}$ is not Laplacian integral whenever $c \geq 4a^2 - 2a$.*

Proof. Use the complement quintic $h = h_{a,b,c}$ of Section 5. Its first endpoint is the cubic

$$h(1) = Lc^3 + Qc^2 + Rc + S, \tag{10}$$

where

$$\begin{aligned} L &= a^2 + b^2 - 1, \\ Q &= a^3 - 4a^2b^2 + 2a^2b - a^2 + 2ab^2 + ab - 3a + b^3 - b^2 - 3b + 3, \\ R &= (a - 1)(b - 1)(2ab + 3a + 3b - 3), \\ S &= (a - 1)(b - 1)(a + b - 1)(ab + a + b - 1). \end{aligned}$$

Here $L, R, S > 0$. In fact the sharper condition $Lc + Q \geq 0$ already gives $h(1) > 0$. Put $T(a) = 4a^2 - 2a$. Then

$$Q + T(a)L = b^2(b-1) + (2a^2 + a - 3)b + K(a),$$

where $K(a) = 4a^4 - a^3 - 5a^2 - a + 3$. For $a = 2 + u$, $u \geq 0$,

$$K(2+u) = 4u^4 + 31u^3 + 85u^2 + 95u + 37 > 0.$$

Consequently $Lc + Q > 0$ for $c \geq T(a)$. Since $h(0) < 0$, a root lies in $(0, 1)$, producing a noninteger graph eigenvalue in $(V - 1, V)$ by (9) and Lemma 2.1. □

Theorem 5.4. *For distinct primes p, q, r and all $b, c \geq 1$, the graph $G_{p^3q^br^c}$ is not Laplacian integral. Consequently every three-prime exponent vector with minimum entry at most three gives a nonintegral graph.*

Proof. Theorems 5.1, 5.2 and 4.7 leave $4 \leq b < c$. Proposition 5.3 settles $c \geq 30$. At $a = 3$, equation (10) is

$$\begin{aligned} h_{3,b,c}(1) = (b^2+8)c^3 + (b-1)(b^2 - 30b - 12)c^2 \\ + 6(b-1)(3b+2)c + 4(b-1)(b+2)(2b+1). \end{aligned}$$

The remaining domain $4 \leq b < c < 30$ has exactly $\binom{26}{2} = 325$ pairs. Exhaustive integer evaluation of this cubic gives 257 positive and 68 negative values, with no zero. Table 1 lists every negative range; all other pairs in this finite domain have positive values. The complete polynomial and separate Horner sign certificates are in `results/distinct-tail.json`; their construction is described in Section A. Thus a positive value gives a root in $(0, 1)$.

b	4	5	6	7	8	9	10	11	12	13
$c_{\min}$	5	6	7	8	9	10	11	12	13	14
$c_{\max}$	13	15	16	17	17	16	16	15	14	14

Table 1: Every negative $h_{3,b,c}(1)$ in $4 \leq b < c < 30$; each column includes all integers between its two endpoints.

For the negative values, the second endpoint is positive except at $(b, c) = (4, 5)$. Indeed, for $v \geq 0$,

$$h_{3,4,6+v}(2) = 104v^3 + 1238v^2 + 4018v + 2344 > 0.$$

For $b \geq 5$, write $b = 5 + u$, $c = 6 + u + v$, with $u, v \geq 0$. Direct expansion gives

$$\begin{aligned} h_{3,5+u,6+u+v}(2) = (5u^2 + 54u + 153)v^3 \\ + (20u^3 + 262u^2 + 1179u + 1863)v^2 \\ + (25u^4 + 426u^3 + 2642u^2 + 7011u + 6624)v \\ + 10u^5 + 218u^4 + 1828u^3 + 7200u^2 + 12600u + 6480 > 0. \end{aligned}$$

These regions exhaust $4 \leq b < c$ except $(4, 5)$. Whenever $h(1) < 0 < h(2)$, a root lies in $(1, 2)$, giving a noninteger graph eigenvalue in $(V - 2, V - 1)$.

Finally

$$h_{3,4,5}(x) = x^5 - 83x^4 + 2327x^3 - 24133x^2 + 60576x - 42480.$$

Here $h(1) = -3792$ and $h(3/2) = 48825/32 > 0$. The root in $(1, 3/2)$ lifts to a graph eigenvalue in $(233/2, 117)$, since $V = 118$. Every interval used contains no integer. The finite nonvanishing certificate and the written infinite-range arguments exhaust all b, c; Theorems 5.1 and 5.2 give the final assertion. □

6 Small-minimum certificates and a uniform low root

The complement quotient also makes sense for positive real support weights. With $W = \mathrm{diag}(w)$ and disjointness edges on the six supports,

$$z^T W C z = \sum_{\{S,T\}:\, S\cap T=\varnothing} w_S w_T (z_S - z_T)^2.$$

Thus $W^{1/2} C W^{-1/2}$ is positive semidefinite. The singleton triangle and the three complementary-pair edges connect this support graph, so its kernel is one-dimensional. This justifies the real-parameter spectral arguments below as well as their integer specializations.

The uniform cutoff leaves a complete finite domain at minima four through seven. A symmetric Schur complement then supplies a positive root below three at every larger minimum, including when an endpoint sign change misses a pair of low roots.

6.1 The complete minimum-through-seven certificate

The uniform cutoff makes the requested diagnostic range complete for each of the minimum exponents $4, 5, 6, 7$. Here the relevant quintic is $g_B(x) = \det(xI - B)/x$ for H, whereas the preceding endpoint arguments use the complement quintic $h_C(x) = -g_B(N - x)$.

Corollary 6.1. *Every positive three-prime exponent triple with minimum entry at most seven gives a nonintegral ideal intersection graph.*

Proof. Theorem 5.4 handles minimum entries at most three. For a minimum entry $a \in \{4, 5, 6, 7\}$, repeated entries are covered by Theorem 4.7, and Proposition 5.3 settles $c \geq 4a^2 - 2a$. The remaining ordered triples lie in the exact finite domain $a < b < c < 4a^2 - 2a$.

For all 27 562 triples in this domain, the integer discriminant $\Delta_{a,b,c}$ of g_B is positive and strictly between two consecutive integer squares. Table 2 gives the complete counts. The certificate `results/open-region-discriminants.csv` records every triple, discriminant and integer q satisfying $q^2 < \Delta_{a,b,c} < (q+1)^2$. Each discriminant is independently recomputed as a 9×9 integer Sylvester determinant by `scripts/check_open_region_certificate.py`, using fraction-free elimination and exact divisions. The certificate exhausts the domain without omitted or duplicate triples.

If all roots of the monic quintic g_B were integers, its discriminant, the product of the squared pairwise root differences, would be an integer square. Thus every certified quintic has a noninteger root. The quotient is similar to a real symmetric Laplacian, so the root is real and nonzero; Lemma 2.1 gives a noninteger graph eigenvalue. This complete finite certificate and the written cutoff exhaust all triples with minimum entry at most seven. □

6.2 The symmetric Schur complement

Proposition 6.2. *Order the exponents as $a \leq b \leq c$, and let m be an integer with $2 \leq m < a$. If*

$$(c - m)(a + b + c - m) < mab, \tag{11}$$

Minimum a	4	5	6	7	Total
Exclusive cutoff	56	90	132	182	—
Complete pairs	1275	3486	7750	15051	27562
Nonsquare discriminants	1275	3486	7750	15051	27562

Table 2: Exhaustive discriminant certificates in the region derived from the uniform cutoff.

then the six-support complement quotient has exactly two nonzero eigenvalues in $(0, m)$*. If, in addition,*

$$(a - m + 1)(a + b + c - m + 1) > (m - 1)bc, \tag{12}$$

both lie in $(m - 1, m)$*, so* $G_{p^a q^b r^c}$ *is not Laplacian integral.*

Proof. Put $s = a+b+c$ and $P = abc$. Symmetrize C by the square roots of its cell sizes and reorder the classes as $1, 2, 3, 23, 13, 12$. The resulting real symmetric matrix is

$$\mathcal{C} = \begin{pmatrix} A & -\sqrt{P}I \\ -\sqrt{P}I & D \end{pmatrix}, \quad D = \operatorname{diag}(a, b, c),$$

where A has diagonal $(b + c + bc, a + c + ac, a + b + ab)$ and off-diagonal entries $A_{ij} = -\sqrt{a_i a_j}$, with $(a_1, a_2, a_3) = (a, b, c)$. For $0 < x < a$, the block $D - xI$ is positive definite, and Sylvester's law of inertia gives

$$\operatorname{inertia}(\mathcal{C} - xI) = \operatorname{inertia}(D - xI) + \operatorname{inertia}(K(x)),$$

where

$$K(x) = \operatorname{diag}(k_a(x), k_b(x), k_c(x)) - vv^T, \quad v = (\sqrt{a}, \sqrt{b}, \sqrt{c})^T, \quad k_t(x) = s - x - \frac{xP}{t(t - x)}.$$

For fixed $x > 0$, the function $k_t(x)$ increases with $t > x$, since

$$\frac{\partial k_t(x)}{\partial t} = \frac{xP(2t - x)}{t^2(t - x)^2} > 0.$$

Condition (11) is exactly $k_c(m) < 0$. Thus all three diagonal values are negative, so $K(m)$ is negative definite. The matrix $\mathcal{C} - mI$ has three negative and three positive eigenvalues. The complement support graph is a triangle with one pendant cell at each vertex; it is connected, so C has a simple zero eigenvalue and all other eigenvalues are positive. Exactly two nonzero eigenvalues therefore lie in $(0, m)$.

For $x = m - 1$, condition (12) says $k_a(x) > 0$, so all three $k_t(x)$ are positive. Congruence by the inverse square roots of these values turns $K(x)$ into $I - zz^T$, where

$$\|z\|^2 = \frac{a}{k_a(x)} + \frac{b}{k_b(x)} + \frac{c}{k_c(x)} > 1,$$

because each $k_t(x) < s$. Consequently $K(x)$ has one negative and two positive eigenvalues, and the only eigenvalue of C below x is zero. Both low positive eigenvalues lie in $(m - 1, m)$. They are less than $a < N$, so the complement identity and universal-vertex lift apply. Their graph eigenvalues lie in $(V - m, V - m + 1)$, which contains no integer. □

6.3 The uniform comparison at three

Theorem 6.3. *For every ordered real triple $0 < a \le b \le c$, the six-support complement quotient has a nonzero eigenvalue in $(0, 3)$.*

Proof. Use $s = a + b + c$, $P = abc$ and $v = (\sqrt{a}, \sqrt{b}, \sqrt{c})^T$ from Proposition 6.2, and define

$$L = \operatorname{diag}\left(s - \frac{3P}{a^2}, s - \frac{3P}{b^2}, s - \frac{3P}{c^2}\right) - vv^T.$$

Expansion of this diagonal-minus-rank-one determinant gives

$$\det L = 6\big(c(a-b)^2 + b(a-c)^2 + a(b-c)^2\big),$$
$$v^T L v = -3P\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) < 0.$$

The determinant and Rayleigh value give exactly two negative eigenvalues at unequal parameters. At equal parameters $L = -vv^T$; thus L is nonpositive on a two-dimensional subspace in both cases.

For $a > 3$, the identity

$$K(3) = L - \operatorname{diag}_{t\in\{a,b,c\}}\left(3 + \frac{9P}{t^2(t-3)}\right)$$

subtracts a positive definite matrix. Since $D - 3I > 0$, this gives at least two negative eigenvalues of $\mathcal{C} - 3I$.

If $b \le 3$, restrict $\mathcal{C} - 3I$ to the first two singleton supports and their paired supports. Its blocks are $\left[\begin{smallmatrix} A_0 & -\sqrt{P}I_2 \\ -\sqrt{P}I_2 & D_0 \end{smallmatrix}\right]$ with $D_0 \le 0$. On vectors (z, tz), the quadratic form is at most $z^T(A_0 - 2t\sqrt{P}I_2)z$, which is negative definite for large t.

It remains to consider $a \le 3 < b$. Eliminate the positive paired diagonals $b - 3, c - 3$. On singleton supports $2, 3$, the vector $(\sqrt{b}, \sqrt{c})$ has quadratic value

$$q = (a-3)(b+c) - 3P\left(\frac{1}{b-3} + \frac{1}{c-3}\right) < 0.$$

Compress the remaining matrix to singleton 1, its paired support, and this vector. The resulting matrix is

$$\begin{pmatrix} M & -\sqrt{P} & r \\ -\sqrt{P} & a-3 & 0 \\ r & 0 & q \end{pmatrix}, \quad M = bc + b + c - 3 > 0, \quad r = -\sqrt{a}(b+c).$$

Its determinant $M(a-3)q - Pq - (a-3)r^2$ is positive. Its positive and negative diagonal entries therefore force exactly two negative eigenvalues. This includes the singular boundary $a = 3$.

In every case $\mathcal{C} - 3I$ has at least two negative eigenvalues. Since $\mathcal{C}$ is positive semidefinite with a simple zero, at least one positive eigenvalue lies strictly below three. □

7 The endpoint-one sum bound and spectral separation

On endpoint one the sum of the two larger exponents is quadratic in the minimum. We retain the full spectral statement and its written proof; the sum bound and the symmetric quotient are the inputs used in the largest-root completion.

Theorem 7.1. *Let $2 \le a \le b \le c$ be real and put $s = a + b + c$. The ordered eigenvalues of the six-support complement quotient, counted with multiplicity, satisfy*

$$0 = \lambda_1 < \lambda_2 \le \lambda_3 < a + b < s \le \lambda_4 \le \lambda_5 \le \lambda_6.$$

Equality $\lambda_4 = s$ holds exactly at $(a, b, c) = (2, 2, 2)$. If also $h_C(1) = 0$, then

$$b + c \ge 2a^2 - 2a + 1,$$

with equality exactly when $b = a$ and $c = (a - 1)(2a - 1)$. If in addition $a > 2 + \sqrt{3}$, root one is simple and is the smallest positive quotient root; all four other quotient roots are strictly greater than a. Their ordered roots satisfy

$$a < \mu_1 < \min\{c, a + b\}, \qquad s < \mu_2 \le \mu_3 \le \mu_4.$$

Proof. Let $S = W^{1/2} C W^{-1/2}$, with $W = \operatorname{diag}(a, b, c, ab, ac, bc)$. It is symmetric positive semidefinite with a one-dimensional zero kernel. The principal block on supports $3, 12, 13, 23$ is

$$\begin{pmatrix} ab + a + b & -\sqrt{abc} \\ -\sqrt{abc} & c \end{pmatrix} \oplus (b) \oplus (a).$$

The two-dimensional block has polynomial $Q(x) = x^2 - (ab + a + b + c)x + c(a + b)$, with $Q(a + b) = -ab(a + b) < 0$. Exactly three eigenvalues of this four-dimensional block lie below $a + b$; interlacing gives $\lambda_3 < a + b$.

The principal block on singleton supports $1, 2$, shifted by s, is

$$\begin{pmatrix} bc - a & -\sqrt{ab} \\ -\sqrt{ab} & ac - b \end{pmatrix}.$$

Its first leading minor is $bc - a \ge a(a - 1) > 0$. Its determinant is $c(abc - a^2 - b^2)$, where

$$abc - a^2 - b^2 = ab(c - b) + (a - 1)(b - a)(b + a) + (a - 2)a^2.$$

This is positive except at $(2, 2, 2)$. Away from that point, interlacing gives $\lambda_5, \lambda_6 > s$. Put $p = ab + ac + bc$, $r = abc$. Direct expansion gives $h_C(s) = -rsV$, where

$$V = r(s + 3) - sp = a^2(bc - b - c) + b^2(ac - a - c) + c^2(ab - a - b) > 0.$$

The three brackets are nonnegative and vanish simultaneously only at $(2, 2, 2)$. Thus no eigenvalue equals s. At least three lie below it and at most four. The negative determinant $sh_C(s)$ requires an odd number below it, counting multiplicity; hence exactly three lie below it and $\lambda_4 > s$. At the boundary the characteristic polynomial is $x(x - 6)(x^2 - 12x + 12)^2$, giving $\lambda_4 = s = 6$.

Now suppose $h_C(1) = 0$. Put $q = b + c$, $z = bc$ and $K = 4a^2 - 2a + 1$. The generic determinant is

$$\begin{aligned} h_C(1) &= (q - K)z^2 - a(a - 1)(q + 2a - 1)z + P_a(q), \\ P_a(q) &= (a^2 - 1)q^3 + (a^3 - a^2 - 3a + 3)q^2 - 3(a - 1)^2 q - (a - 1)^3. \end{aligned}$$

Since $(b - a)(c - a) \ge 0$, we have $z \ge z_0 = a(q - a)$. When $q \le Q := 2a^2 - 2a + 1$, the coefficient $q - K \le -2a^2$ is negative, and the difference between the displayed polynomial at z and at z_0 is

$$(z - z_0)\big[(q - K)(z + z_0) - a(a - 1)(q + 2a - 1)\big] \le 0,$$

strictly unless $z = z_0$. Set $t = q - a \geq a$. At z_0 the value factors as

$$\big[(a-1)(2a-1) - t\big]B(a,t),$$

where $B = a^3 - 2a^2t^2 + a^2t + a^2 + 3at - 3a + t^2 - 2t + 1$. For $t = a + u$, with $u \geq 0$, the identity

$$\begin{aligned} -B(a, a+u) = {} & (2a^2-1)u^2 \\ & + \big[(a-2)(4a^2+7a+9) + 20\big]u \\ & + (a-2)\big[2a^3 + a(2a-1) + 3\big] + 5 \end{aligned}$$

has strictly positive coefficients for every $a \geq 2$, so $B < 0$. Thus $q < Q$ gives $h_C(1) < 0$. At $q = Q$, equality requires $z = z_0$, hence $b = a$ by ordering, and $c = (a-1)(2a-1)$. The factorization also verifies this equality case.

Now assume also $a > 2 + \sqrt{3}$. The pair-support principal block is $\mathrm{diag}(c, b, a)$. The sum bound implies $b + c - (a^2 + 2a) \geq a^2 - 4a + 1 > 0$, so

$$h_C(a) = -a^2b^2c^2(a^2 + 2a - b - c) > 0.$$

No eigenvalue equals a. The positive determinant $\det(aI - S) = ah_C(a)$ forces an even number of eigenvalues below a, counted with multiplicity. There are at least two, zero and one, and at most three by interlacing. There are therefore exactly two: zero and a simple one. All four remaining roots exceed a.

The pair-support block gives $\lambda_3 \leq c$. The identity $h_C(c) = -a^2b^2c^2(c^2 + 2c - a - b) < 0$ excludes equality. Together with the global total-sum gap, this proves the residual bounds.

The real cutoff is sharp. On the ordered endpoint-one family $b = a$, $c = (a-1)(2a-1)$, we have $h_C(a) = a^4(a-1)^2(2a-1)^2(a^2 - 4a + 1)$. At $a = 2 + \sqrt{3}$ a residual root equals a. For $2 \leq a < 2 + \sqrt{3}$, the determinant at a is negative, so the same interlacing and parity argument gives three roots below a; a residual root lies below a. □

8 The endpoint-two alternative

For integer exponents, a noninteger quotient root μ gives the graph eigenvalue $N + abc - 1 - \mu$ by Lemma 2.1, since $N - \mu \neq 0$. The earlier small-minimum obstructions already lie in the six-support space; the repeated-entry invariant blocks also lie there, with their roots reflected from B to C by (9).

Two geometric bounds confine the endpoint-two surface to a rectangle where positive polynomial identities give a second noninteger root at large minima. A complete finite base handles the smaller distinct triples.

Lemma 8.1. *Put $\alpha = (7 + \sqrt{13})/3$. For ordered real exponents $\alpha \leq a \leq b \leq c$, the equation $h_C(2) = 0$ requires $b \leq 2a - 2$, with equality exactly at $a = \alpha$, $b = c = 2\alpha - 2$. In particular, $b < 2a - 2$ if $a > \alpha$. If $a \geq 8$, it also requires $c < 9a/4 - 8$.*

Proof. First put $b = 2a - 2 + u$ and $c = b + v$, where $u, v \geq 0$. Define

$$\begin{aligned} A_0(u) &= 2u^2 + 7(a-2)u + 2(3a^2 - 14a + 12), \\ B_0(u) &= (a+2)u^3 + (4a^2 + 10a - 12)u^2 \\ &\quad + (4a^3 + 25a^2 - 48a + 12)u + 2(13a^3 - 28a^2 + 8a + 8), \\ D_0(u) &= 2(a-2)(9a^3 + 24a^2 - 56a + 16) \\ &\quad + (33a^3 + 20a^2 - 208a + 136)u + (20a^2 + 23a - 62)u^2 + 4(a+2)u^3, \\ E_0(u) &= 2(3a^3 - a^2 - 8a + 4) + (5a^2 + 3a - 10)u + (a+2)u^2. \end{aligned}$$

Direct expansion of the quotient determinant gives

$$h_C(2) = A_0B_0 + \tfrac{1}{2}(A_0B_0)'v + D_0v^2 + E_0v^3,$$

where the derivative is in u. At $a = 3 + m$, all coefficients of B_0, D_0, E_0 are positive for $m \geq 0$. For $a \geq \alpha$, $A_0(0) = 6(a - \alpha)(a - (7 - \sqrt{13})/3) \geq 0$ and $A_0'(u) = 4u + 7(a - 2) > 0$. Hence $A_0 \geq 0$, vanishing only at $a = \alpha, u = 0$, and $(A_0B_0)' > 0$. The identity is positive unless $a = \alpha, u = v = 0$, where its zero proves the equality case and sharpness.

For $a \geq 8$ and $a \leq b < 2a - 2$, parameterize the proposed maximum tail by

$$a = 8 + m, \qquad b = \frac{a + (2a - 2)t}{1 + t}, \qquad c = \frac{9a}{4} - 8 + v, \qquad m, t, v \geq 0.$$

The inverse $t = (b - a)/(2a - 2 - b)$ has positive denominator. The exact identity is

$$\begin{aligned} 64(1 + t)^3 h_C(2) = {} & 9m^6(1 + 2t)\big[14(1 - t)^2 + 5t + t^2\big] \\ & + 308m^5t(t - 1)^2 + P(m, t, v). \end{aligned}$$

The first two terms are nonnegative. The complete polynomial P, listed in Appendix B, has 84 strictly positive coefficients and constant 3 014 656. Hence the right side is positive, and so is $h_C(2)$. Together with the first bound, this proves $c < 9a/4 - 8$ on the zero surface. □

Theorem 8.2. *For ordered real $40 \leq a \leq b \leq c$ with $h_C(2) = 0$,*

$$h_C(4) > 0 > h_C(5).$$

Consequently the quotient has a noninteger root in $(4, 5)$. For every positive integer exponent triple with $h_C(2) = 0$, the quotient spectrum is nonintegral. The integer conclusion uses the complete finite base specified below and the earlier small-minimum certificates.

Proof. Write $s = a + b + c$, $p = ab + ac + bc$ and $r = abc$, and set

$$D_4 = h_C(4) - 3h_C(2), \qquad D_5 = 4h_C(2) - h_C(5).$$

The full identities in Appendix B give the following positive expansions. For $a = 8 + m, b = 8 + t, c = 8 + u$, D_5 has 44 strictly positive coefficients and constant 4 629 843. Thus $D_5 > 0$ for arbitrary $a, b, c \geq 8$.

Lemma 8.1 confines an endpoint-two zero at $a \geq 40$ to $a \leq b < 2a$ and $a \leq c < 9a/4$. Put

$$a = 40 + m, \qquad b = \frac{a(1 + 2t)}{1 + t}, \qquad c = \frac{a(4 + 9u)}{4(1 + u)}, \qquad m, t, u \geq 0.$$

The inverse substitutions are $t = (b - a)/(2a - b)$ and $u = 4(c - a)/(9a - 4c)$, with positive denominators. The polynomial $64(1 + t)^3(1 + u)^3D_4$ has 112 strictly positive coefficients and constant 611 305 472, all displayed in the appendix. Therefore $D_4 > 0$. At $h_C(2) = 0$ these two signs give $h_C(4) > 0 > h_C(5)$; the intermediate value theorem supplies the root in $(4, 5)$.

For positive integer triples, repeated exponents are nonintegral by Theorem 4.7, and minima at most seven by Corollary 6.1. The only remaining base is

$$8 \leq a \leq 39, \qquad a < b < 2a - 2, \qquad b < c < \frac{9a}{4} - 8.$$

The largest permitted integer c is $\lfloor (9a-33)/4 \rfloor$. This domain contains exactly 8 658 triples; its per-minimum counts are listed in Appendix B. Every endpoint value is nonzero, with smallest absolute value 8 208. The complete integer certificate records $h_C(2)$ by integer Horner evaluation and $\det(2I - C) = 2h_C(2)$ by an independent fraction-free 6×6 Bareiss determinant for every triple. Thus no fully distinct endpoint-two zero occurs in this necessary base. The real tail covers all larger minima, completing the integer conclusion. The proof does not require emptiness of the unbounded integer zero surface. □

9 Completion of the three-prime classification

The largest quotient root supplies the final obstruction. Its open unit interval completes endpoint one without deciding the quartic's smallest root or enumerating integer exponent points on the endpoint surface.

Theorem 9.1. *Let $2 \le a \le b \le c$ be real, with $b - a \ge 3$ and $c \ge a^2 - a$. Put $s = a + b + c$ and $M = bc + s$. Then*

$$M < \lambda_{\max}(C) < M + 1.$$

For integer exponents in this domain, the ideal intersection graph is Laplacian nonintegral. The real interval has a written proof with no finite exponent base.

Proof. The quotient determinant gives

$$h_C(M) = -a^2 bc\, T(a, b, c),$$

where

$$T = (2b^2 - ab + b - a)c^2 - [(a-1)b^2 + a^2]c - ab(a+b).$$

The quadratic coefficient of T in c is positive, and

$$T(a, b, b) = 2b[(b - a + 1)b^2 - ab - a^2] \ge 2b^3(b - a - 1) > 0, \qquad \partial_c T(a, b, b) \ge b^3 > 0.$$

The derivative bound follows from $\partial_c T(a, b, b) = b^2(4b - 3a + 3) - 2ab - a^2$ and $a \le b$. Thus $T > 0$ for $c \ge b$. Put $P(a, b, c) = h_C(M + 1)$. For $a \ge 3$, under $a = 3 + m$, $b = a + 3 + u$, $c = a^2 - a + v$, with $m, u, v \ge 0$, the complete identities in Appendix C express T and P as polynomials with respectively 36 and 170 strictly positive coefficients, including constants 1404 and 513220. For $2 \le a \le 3$, write $a = 2 + m$, $b = a + 3 + u$, $c = b + v$, where $0 \le m \le 1$ and $u, v \ge 0$. The same appendix gives $P \ge B(u, v) > 0$, with all 35 coefficients of B displayed. Thus $h_C(M) < 0 < h_C(M + 1)$, giving a root in $(M, M + 1)$.

To identify it as the largest root, use the symmetric similarity S from Theorem 7.1. For $x > c$, the pair-support block of $xI - S$ is positive definite. Its Schur complement is

$$H(x) = \operatorname{diag}(\ell_a, \ell_b, \ell_c) + zz^T, \qquad z = (\sqrt{a}, \sqrt{b}, \sqrt{c})^T, \qquad \ell_t = x - s - \frac{xabc}{t(x - t)}.$$

At $x = M + 1$, both $x - b$ and $x - c$ exceed bc. Hence

$$\ell_b = c(b - a) + 1 - \frac{abc}{x - b} > c(b - a) + 1 - a > 0,$$
$$\ell_c = b(c - a) + 1 - \frac{abc}{x - c} > b(c - a) + 1 - a > 0.$$

Here $c \geq b \geq a+3$. The b, c principal block of H is positive definite, and $\det(xI - S) = xP > 0$ gives $\det H > 0$. Its remaining scalar Schur complement is therefore positive. Consequently $xI - S$ is positive definite, and every eigenvalue is less than $M+1$. Since $\det(MI - S) = Mh_C(M) < 0$, not all eigenvalues can lie below M, so the largest lies in the stated interval. For integer exponents, the established complement and universal-class lifts transfer its nonintegrality to the ideal intersection graph. □

The first two middle gaps use exponent-root brackets rather than the largest-root interval. The following proof records their signs explicitly so that the classification does not rely on an unstated uniqueness claim.

Lemma 9.2. *Let $a \geq 2$ be real, $d \in \{1, 2\}$ and $b = a + d$. Put $L_1 = 2a^2 - a - 2$ and $L_2 = 2a^2 + a - 6$. As a polynomial in c, $h_C(1)$ has exactly one root above b, lying in $(L_d, L_d + 1)$. Consequently, for integer $a \geq 2$, no integer $c > b$ satisfies $h_C(1) = 0$.*

Proof. For fixed a, b, write $Q(c) = h_C(1) = Ac^3 + Bc^2 + Dc + E$, where

$$\begin{aligned} A &= a^2 + b^2 - 1, \\ B &= a^3 - 4a^2b^2 + 2a^2b - a^2 + 2ab^2 + ab - 3a + b^3 - b^2 - 3b + 3, \\ D &= (a-1)(b-1)(2ab + 3a + 3b - 3), \\ E &= (a-1)(b-1)(a+b-1)(ab + a + b - 1). \end{aligned}$$

Both A and E are positive. Put $L_1 = 2a^2 - a - 2$ and $L_2 = 2a^2 + a - 6$. Direct expansion gives

$$\begin{aligned} -Q(L_1) &= 4a(a+1)(a^4 - 4a^2 - 2a + 6), \\ Q(L_1 + 1) &= 4a(a-1)(a+1)(a^3 - a^2 + 1), \\ -Q(L_2) &= 16a^5 + 72a^4 - 76a^3 - 433a^2 + 69a + 595, \\ Q(L_2 + 1) &= 4(a+2)(2a^5 - 2a^4 - 17a^3 + 25a^2 + 28a - 42), \end{aligned}$$

where each pair uses its own $b = a + d$. The complete positive vectors at $a = 2 + m$, including those for $-Q(b)$, are in Appendix C. Thus $Q(0) > 0 > Q(b)$ and $Q(L_d) < 0 < Q(L_d + 1)$. Moreover,

$$L_1 - b = 1 + 6(a-2) + 2(a-2)^2 > 0, \qquad L_2 - b = 2(a-2)(a+2) \geq 0.$$

The second equality occurs only at $a = 2$; the final open interval still lies strictly above b there. The cubic has one root in each of $(-\infty, 0)$, $(0, b)$ and $(L_d, L_d + 1)$. These three distinct roots exhaust its degree. The unique root above b is therefore in $(L_d, L_d + 1)$. For integer a, the endpoints are consecutive integers, proving the nonvanishing conclusion without a finite base. □

Theorem 9.3 (Three-prime nonintegrality)**.** *For distinct primes p, q, r and positive integers a, b, c, the ideal intersection graph of $\mathbb{Z}_{p^a q^b r^c}$ is Laplacian nonintegral. This global conclusion combines written proofs with the established small-minimum and endpoint-two finite certificates.*

Proof. Sort the exponents. Repeated entries are covered by Theorem 4.7; minima at most seven by Corollary 6.1. Suppose $8 \leq a < b < c$ and the spectrum is integral. Theorem 6.3 supplies a positive quotient root in $(0, 3)$, necessarily one or two. Theorem 8.2 excludes the second alternative, so $h_C(1) = 0$.

For $b - a = 1$ or 2, Lemma 9.2 contradicts integer $c > b$. Otherwise $b - a \geq 3$, and Theorem 7.1 gives $b + c \geq 2a^2 - 2a + 1$. Since $c \geq b$, we have $c > a^2 - a$. Theorem 9.1 supplies a noninteger quotient root, again contradicting integrality. These cases exhaust all positive exponent triples.

The minimum-through-seven theorem retains its 27 562-triple base and earlier small-minimum dependencies; the endpoint-two theorem retains its 8 658-triple base. The new largest-root and small-gap steps have written unbounded proofs. No additional exponent scan or integer-point enumeration is used in this completion. □

10 Further questions

The three-prime classification is complete. The general nonsquarefree problem with four or more distinct prime factors remains open outside the established families in the supporting package. One structural question is whether a quotient restriction or variational argument can handle fully distinct higher-prime exponents without a separate expansion for each repeated value. Effective integer-point enumeration on the endpoint-one surface is another arithmetic question, independent of the nonintegrality theorem proved here.

Declaration of AI use

During this research, DeepSeek was used for diagnostic-script code generation, counterexample analysis and language editing. OpenAI Codex was used for mathematical exploration and proof development, code generation, exact-verification diagnostics and manuscript editing. OpenAI ChatGPT (GPT-6 Pro) was used for an additional review of proof dependencies and presentation. The authors are responsible for the mathematical claims, code and final text; these tools are not authors. The verification scope is specified in Appendix A.

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A Exact evidence for the classification

The computer-assisted classification combines written unbounded proofs with finite bases exhausting the domains derived by those proofs. The repository [17] retains exact integer and rational checkers for Python 3.10 or later and SymPy 1.14.0, without a Lean formalization. Reproduction commands, fixed source and data revisions, and expected results are in `notes/classification-certificate-manifest.md`.

A.1 Repeated entries and small minima

The exact divisor reduction exhausts the first three nonboundary factor indices by all divisors of $24, 108, 320$. The eight positive-b cubics have complete root-free modular certificates; later indices use the written cubic interval. The saved data are in `results/repeated-exponent-completion.json` and the earlier factor-index certificates. At minimum three, the proved cutoff leaves exactly 325 pairs $4 \le b < c < 30$. The first endpoint has 257 positive and 68 negative values with no zero. The complete polynomial and separate integer-Horner certificate is in `results/distinct-tail.json`.

At minima four through seven, the complete domain $a < b < c < 4a^2 - 2a$ has 27 562 triples. Each row in `results/open-region-discriminants.csv` records the integer discriminant and consecutive-square brackets. Independent fraction-free 9×9 Sylvester determinants, exact divisions, completeness and absence of duplicate triples are checked by `scripts/check_open_region_certificate.py`; its saved report is `results/open-region-certificate-check.json`. These finite inputs remain part of Corollary 6.1.

A.2 Endpoint-two completion

Appendix B displays the complete 84-, 44- and 112-term positive identities, the two nonnegative square expressions and every per-minimum count. `scripts/verify_endpoint_two_middle_cutoff.py` checks the cubic identity, the 36 positive coefficients at $a = 3+m$ and the sharp boundary.

The necessary base at minima eight through thirty-nine contains exactly 8 658 fully distinct triples. Integer Horner and 6×6 Bareiss evaluations agree and give no endpoint zeros; the revision-pinned checker and its complete CSV and JSON preserve this evidence. `scripts/verify_endpoint_two_coefficients.py` checks all 240 displayed coefficients and 32 domain counts by direct determinant and an arithmetic formula.

A.3 Written spectral bounds

The uniform positive-root bound below three holds for all positive real parameters; `scripts/verify_uniform_low_root_positive.py` checks its principal-block identities, singular boundary and sharp limiting family. The sum comparison uses the displayed quadratic with three positive coefficients. Its residual-root separation has the sharp real cutoff $a > 2 + \sqrt{3}$; `scripts/verify_residual_spectrum_cutoff.py` checks the anchor and sharpness identities. The largest-root interval uses the positive 36- and 170-term identities for $a \geq 3$, a 35-coefficient positive lower bound for $2 \leq a \leq 3$, and a symmetric Schur complement. The two small-gap proofs use six complete sign vectors at $a = 2 + m$, proving the brackets for all real $a \geq 2$. Appendix C retains every coefficient. The revision-pinned largest-root checker checks the quotient, Schur determinant, original tables and historical small-gap vectors. `scripts/verify_largest_root_coefficients.py` checks the current tables directly; `scripts/verify_largest_root_minimum_two.py` checks the slab bound. Fixed controls supplement the written proofs. The finite bases above remain essential; separate minimum-eight and middle-at-most-fifteen certificates are unnecessary for this route.

`scripts/verify_proof_structure.py` reconstructs the balanced factorization and endpoint-one cubic directly from the quotient. It checks the rescaling, nonzero implicit-function derivative and every displayed fixed-gap coefficient in Section 3. The supplementary fixed-gap asymptotic is a written analytic result; the classification uses the exact small-gap brackets and largest-root interval already described.

B Complete endpoint-two positive identities and finite base

This appendix supplies all coefficients used in Lemma 8.1 and Theorem 8.2. A vector $[v_0, \dots, v_k]$ represents $\sum_{\ell=0}^{k} v_\ell m^\ell$. The identities are coefficient identities for the generic quotient, rather than values sampled at exponent triples.

B.1 Maximum-bound residual

For $a = 8 + m$, $b = [a + (2a - 2)t]/(1 + t)$ and $c = 9a/4 - 8 + v$, define P by

$$\begin{aligned} 64(1+t)^3 h_C(2) = {} & 9m^6(1+2t)\big[14(1-t)^2 + 5t + t^2\big] \\ & + 308m^5 t(t-1)^2 + P(m,t,v). \end{aligned}$$

The complete expansion $P = \sum_{i,j=0}^{3} P_{ij}(m) t^i v^j$ is:

i	j	Coefficients in ascending powers of m
0	0	$[3014656, 4968448, 2878080, 766780, 100060, 6027]$
0	1	$[3489792, 4092928, 1603664, 277392, 21796, 632]$
0	2	$[1124352, 751424, 170816, 16144, 544]$
0	3	$[75520, 27392, 3264, 128]$
1	0	$[393216, 3104768, 2984064, 974712, 131516, 6322]$
1	1	$[7569408, 11273216, 4914464, 907536, 74792, 2252]$
1	2	$[4098048, 2845312, 670016, 65568, 2288]$
1	3	$[308736, 115456, 14208, 576]$
2	0	$[19243008, 14490880, 5200064, 995716, 79824, 1]$
2	1	$[6706176, 11799040, 5473584, 1047616, 88300, 2684]$
2	2	$[5053440, 3608768, 872192, 87472, 3120]$
2	3	$[413952, 158720, 20032, 832]$
3	0	$[44646400, 39516672, 14832512, 2952456, 316272, 15946]$
3	1	$[6137856, 7480320, 3084384, 563904, 46760, 1416]$
3	2	$[2217984, 1597824, 391424, 39840, 1440]$
3	3	$[180736, 70656, 9088, 384]$

There are 84 nonzero coefficients, all strictly positive, and the constant is $3\,014\,656$.

B.2 The sign difference at five

With $s = a + b + c$, $p = ab + ac + bc$ and $r = abc$, direct expansion gives

$$D_5 = 4h_C(2) - h_C(5) = 9s^3 + 12s^2p + 6s^2r + 3sp^2 + 9p^2 + 3pr \\ - 279s^2 - 168sp - 87sr + 1683s + 468p + 9r - 2997.$$

For $a = 8 + m, b = 8 + t, c = 8 + u$, write $D_5 = \sum_{i,j=0}^{3} V_{ij}(m)t^i u^j$. Its full coefficient vectors are:

i	j	Coefficients in ascending powers of m
0	0	$[4629843, 996483, 67377, 1353]$
0	1	$[996483, 186366, 10395, 156]$
0	2	$[67377, 10395, 417, 3]$
0	3	$[1353, 156, 3]$
1	0	$[996483, 186366, 10395, 156]$
1	1	$[186366, 29079, 1239, 12]$
1	2	$[10395, 1239, 30]$
1	3	$[156, 12]$
2	0	$[67377, 10395, 417, 3]$
2	1	$[10395, 1239, 30]$
2	2	$[417, 30]$
2	3	$[3]$
3	0	$[1353, 156, 3]$
3	1	$[156, 12]$
3	2	$[3]$
3	3	$[0]$

The zero vector is an absent component. All 44 nonzero coefficients are strictly positive, including constant $4\,629\,843$. This establishes $h_C(5) < 0$ on endpoint two for all exponents at least eight.

B.3 The sign difference at four

The corresponding generic identity is

$$D_4 = h_C(4) - 3h_C(2) = 2r^2 - 2s^2r - 2sp^2 - 2pr - 6s^2p - 4s^3 \\ + 120s^2 + 72sp + 36sr - 624s - 4p^2 - 168p - 4r + 928.$$

For $a = 40{+}m$, $b = a(1{+}2t)/(1{+}t)$ and $c = a(4{+}9u)/(4(1{+}u))$, the polynomial $64(1{+}t)^3(1{+}u)^3 D_4 = \sum_{i,j=0}^{3} U_{ij}(m) t^i u^j$ has the following complete coefficient vectors:

i	j	Coefficients in ascending powers of m
0	0	[611305472, 13438528512, 1669177344, 83080704, 2067840, 25728, 128]
0	1	[225809391616, 101554217216, 10552548352, 490948992, 11793920, 143584, 704]
0	2	[575935266816, 199453648896, 19487672672, 881946672, 20857320, 251384, 1224]
0	3	[218865180672, 94951560192, 9789841664, 453837884, 10879740, 132278, 648]
1	0	[181014296576, 89306490880, 9443545088, 442607616, 10675840, 130304, 640]
1	1	[2226833891328, 644107380480, 59528518784, 2622449728, 61045920, 728192, 3520]
1	2	[5054388092928, 1267215288320, 110831526304, 4741723008, 108398120, 1277272, 6120]
1	3	[2483032498176, 647063198720, 57495972608, 2481039396, 57022790, 674384, 3240]
2	0	[440930932736, 161760308224, 16049227264, 731519232, 17370880, 209920, 1024]
2	1	[4630697975808, 1163490925312, 101856239872, 4360110016, 99710080, 1175200, 5632]
2	2	[10376062353408, 2312188925952, 191477361952, 7938026736, 177794520, 2065240, 9792]
2	3	[5505239310336, 1225715108864, 101452569344, 4204474452, 94149320, 1093460, 5184]
3	0	[193009477632, 77502509056, 7857856000, 361629184, 8634112, 104704, 512]
3	1	[2343969284096, 585405371648, 51112650880, 2184643072, 49911296, 587872, 2816]
3	2	[5508276535296, 1195972016128, 97721281760, 4018207392, 89506456, 1035632, 4896]
3	3	[3069924728832, 652290753536, 52684826880, 2150833004, 47677022, 549714, 2592]

All 112 nonzero coefficients are strictly positive, including constant 611 305 472. On the rectangle furnished by Lemma 8.1, this establishes $h_C(4) > 0$ on endpoint two at minimum at least forty.

B.4 Complete necessary integer base

The base is exactly $8 \le a \le 39$, $a < b < 2a - 2$, $b < c < 9a/4 - 8$. Its per-minimum counts are given below; every row has zero endpoint-two zeros. The largest integer c is $\lfloor (9a - 33)/4 \rfloor$.

a	Triples	a	Triples
8	0	24	210
9	3	25	253
10	6	26	276
11	10	27	300
12	15	28	325
13	28	29	377
14	36	30	405
15	45	31	434
16	55	32	464
17	78	33	525
18	91	34	558
19	105	35	592
20	120	36	627
21	153	37	697
22	171	38	735
23	190	39	774

The total is 8 658, and the smallest absolute endpoint value is 8 208. Every row of the complete CSV records both integer-Horner $h_C(2)$ and independent 6×6 Bareiss $\det(2I - C) = 2h_C(2)$. The certificate records the CSV hash, source hashes, per-minimum counts and direct checks. The repeated control $(10, 10, 12)$ has $h_C(2) = 0$ and $h_C(4) < 0$; it is outside this fully distinct base and below the real-tail threshold. Its nonintegrality follows from the earlier repeated-exponent theorem. No $(4, 5)$ root bracket is asserted for all smaller minima.

C Positive identities for the largest-root completion

Put $s = a + b + c$, $M = bc + s$ and

$$T = (2b^2 - ab + b - a)c^2 - [(a-1)b^2 + a^2]c - ab(a+b), \qquad P = h_C(M+1).$$

For $a = 3 + m$, $b = a + 3 + u$, $c = a^2 - a + v$, each row below gives the ascending m coefficient vector of $u^i v^j$. All rows and all coefficients are displayed: 36 terms for T, 170 for P. Every coefficient is positive. Thus the identities establish the two signs on the unbounded real domain with $a \geq 3$, rather than by extrapolating from integer controls.

The independent Schur reconstruction uses $L_t = (x - s)(x - t) - xabc/t$ and the polynomial identity

$$\det(xI - C) = L_a L_b L_c + a(x-a)L_b L_c + b(x-b)L_a L_c + c(x-c)L_a L_b.$$

This specifies P directly and agrees with the support-derived determinant.

C.1 The lower sign: T

i	j	Ascending m coefficient vector
0	0	[1404, 3042, 2517, 1036, 224, 24, 1]
0	1	[603, 684, 261, 39, 2]
0	2	[57, 15, 1]
1	0	[603, 1188, 875, 305, 50, 3]
1	1	[240, 240, 72, 6]
1	2	[22, 3]
2	0	[57, 103, 67, 19, 2]
2	1	[22, 19, 4]
2	2	[2]

C.2 The upper sign: P

i	j	Ascending m coefficient vector
0	0	[513220, 2023568, 3509410, 3571354, 2381713, 1095075, 354193, 80470, 12554, 1277, 76, 2]
0	1	[461006, 1356418, 1739823, 1279064, 593441, 179723, 35391, 4351, 302, 9]
0	2	[142258, 295089, 256850, 120963, 33056, 5205, 436, 15]
0	3	[18594, 23290, 11201, 2530, 270, 11]
0	4	[882, 399, 60, 3]
1	0	[461006, 1812246, 3118532, 3129572, 2042989, 911507, 283142, 61012, 8889, 828, 44, 1]
1	1	[398473, 1159066, 1455815, 1037238, 460760, 131613, 23978, 2659, 161, 4]
1	2	[118957, 239834, 199807, 88401, 22137, 3087, 219, 6]
1	3	[15059, 17725, 7741, 1509, 131, 4]
1	4	[693, 253, 29, 1]
2	0	[142258, 547789, 915785, 883535, 547176, 227591, 64328, 12165, 1466, 101, 3]
2	1	[118957, 336158, 404388, 271110, 110621, 27997, 4258, 352, 12]
2	2	[34629, 66544, 51528, 20429, 4311, 450, 18]
2	3	[4285, 4597, 1712, 248, 12]
2	4	[193, 49, 3]
3	0	[18594, 69414, 111456, 101999, 58928, 22337, 5555, 872, 78, 3]
3	1	[15059, 40917, 46523, 28811, 10495, 2240, 257, 12]
3	2	[4285, 7749, 5460, 1864, 303, 18]
3	3	[520, 496, 147, 12]
3	4	[23, 3]
4	0	[882, 3171, 4853, 4172, 2218, 750, 158, 19, 1]
4	1	[693, 1797, 1911, 1075, 339, 57, 4]
4	2	[193, 325, 204, 57, 6]
4	3	[23, 19, 4]
4	4	[1]

C.3 The two small-gap exponent brackets

For $b = a + d$, $d = 1, 2$, put $Q(c) = h_C(1)$, $L_1 = 2a^2 - a - 2$ and $L_2 = 2a^2 + a - 6$. The following rows give the complete ascending m vectors at $a = 2 + m$ for $-Q(b)$, $-Q(L_d)$ and $Q(L_d + 1)$. Their positivity proves every sign used in Lemma 9.2.

d	Expression	Ascending m coefficient vector
1	$-Q(b)$	$[64, 480, 848, 664, 264, 52, 4]$
1	$-Q(L_d)$	$[48, 376, 768, 648, 264, 52, 4]$
1	$Q(L_d + 1)$	$[120, 412, 592, 456, 196, 44, 4]$
2	$-Q(b)$	$[57, 1689, 2513, 1516, 456, 68, 4]$
2	$-Q(L_d)$	$[57, 1009, 2119, 1140, 232, 16]$
2	$Q(L_d + 1)$	$[160, 360, 640, 892, 476, 104, 8]$

C.4 The upper sign for minima between two and three

Put $a = 2+m$, $b = 5+m+u$, $c = b+v$, with $0 \le m \le 1$ and $u, v \ge 0$, and expand $P = \sum p_{kij} m^k u^i v^j$. Define $B = \sum_{i,j} (p_{0ij} + \sum_{k \ge 1,\, p_{kij} < 0} p_{kij}) u^i v^j$. The exact identity

$$P - B = \sum_{k \ge 1,\, p_{kij} > 0} p_{kij} m^k u^i v^j + \sum_{k \ge 1,\, p_{kij} < 0} |p_{kij}| (1 - m^k) u^i v^j \ge 0$$

holds on this slab. The following complete ascending u vectors of v^j give B directly from the determinant above. All 35 coefficients are positive, so $P \ge B \ge 259876 > 0$.

j	Ascending u coefficient vector of B
0	$[259876, 480352, 371193, 157834, 41170, 6792, 693, 40, 1]$
1	$[240176, 371193, 236751, 82340, 16980, 2079, 140, 4]$
2	$[77666, 100317, 52638, 14520, 2226, 180, 6]$
3	$[10700, 11468, 4800, 987, 100, 4]$
4	$[540, 468, 147, 20, 1]$